Aninda Majumdar

Electronics & Communication Engineering Undergraduate
• Embedded Systems • Sensor Interfacing • Hardware Integration • PCB Design • Mechatronics
gmail github linkedin

Education

VIT-AP University 2023 – 2027
Bachelor of Technology in Electronics and Communication Engineering (ECE) • CGPA: 8.02

Technical Skills

Hands-on Hardware: Soldering & wiring, board bring-up and debugging, multi-board system integration

Embedded Systems: Arduino, STM32 Family, ESP32 Family, K210, SG2002, NF Family • C, C++, Python, FreeRTOS

Sensors & Interfacing: IMU (MPU-9250), MEMS microphone arrays (INMP441), GNSS (MAX-M10S), I^2C, SPI, UART, I^2S, 1-Wire, USB, LoRa, Wi-Fi, BLE

Hardware Design: KiCad, Multi-layer PCB Design, Power Electronics (LDO & switching regulators), ESD Protection

Analog: MOSFET, BJT, Op-Amps, Instrumentation Amplifiers

Signal Processing & Modelling: MATLAB, DSP (DFT/FFT, STFT, filtering), librosa, scikit-learn, PyTorch

CAD: Fusion 360 (currently learning), PCB-to-enclosure fit and board mechanical layout in KiCad

Projects

Acoustic Localization & Human Sensing Edge Node **2026**

K210 • ESP32 • SX1262 LoRa • MAX-M10S GNSS • MPU-9250 • INMP441 • KiCad • C • Python

- Designed and integrated a two-board search-and-rescue sensing node: a 2-layer Ø100 mm array board (8× INMP441 I^2S microphones on a 90 mm circle, Kendryte K210, buck regulators) and a 4-layer 50×75 mm main board (ESP32, SX1262 LoRa, MAX-M10S GNSS, MPU-9250 IMU, CH334F USB hub), joined by an 11-pin castellated edge.
- Interfaced multiple sensors over I^2S, I^2C, SPI and UART; wrote K210 firmware in C for phase-coherent 8-channel capture on a shared clock, an on-chip voice detector, and GCC-PHAT direction-of-arrival with cross-channel bias calibration.
- Built Python host tooling: recorder with sequence-number drop detection, dataset QC with a confound check, and an HTTP API with a live radial dashboard.
- GitHub: CA_LA_ES

Wi-Fi CSI Activity Recognition: Dual ESP32-S3 + SG2002 Board **2025 – 2026**

ESP32-S3 • ESP-IDF • FreeRTOS • Sophgo SG2002 • KiCad • Python • PyTorch

- Designed a single 80×80 mm custom board in KiCad integrating dual ESP32-S3 and a Sophgo SG2002 (QFN-88, custom symbol/footprint) over a USB hub, and handled fabrication, assembly and bring-up.
- Wrote ESP-IDF/FreeRTOS firmware streaming Wi-Fi CSI over 921,600-baud UART from the ESP32s to the SG2002 for device-free human activity recognition.
- Deployed a compact EfficientNetV2-based classifier on the SG2002 for on-device inference (model training done with teammates).
- GitHub: CSI_Collection

STM32 8×8 LED Matrix Driver **2026**

STM32F103C8T6 • KiCad • USB-C • CH340G

- Designed a custom 4-layer STM32F103 PCB driving an 8×8 LED matrix using transistor-based multiplexing.
- Designed discrete high-side and low-side BJT driver stages, sized against datasheet current and switching specs.
- Laid out the board with dedicated power/ground planes and ESD protection; caught and corrected crystal load-capacitor values during design review.
- Planned timer-driven multiplexing firmware with framebuffer rendering and Bit Angle Modulation (BAM) brightness control.
- Modelled a custom enclosure for the board in Fusion 360 (hairpin form factor), designing around PCB outline, mounting points and USB-C cutout.
- GitHub: github.com/irocobble/LED_Matrix

Achievements & Leadership

- **Internal Recognition at Smart India Hackathon (SIH)** for an IoT-based embedded systems project.
- Led a KiCad PCB design workshop and two-round competition (schematic/Gerber and live layout) for a college tech fest.
- Oracle Cloud Infrastructure Certified Data Science Professional.